Arista Access Network SOP - Optical Transport Failure Resolution (KR-ACC-56, KR Puram Access)

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_7277

Description

This Standard Operating Procedure (SOP) outlines the steps to be taken in the event of an Optical Transport Failure on the Arista Access Node (Node ID: KR-ACC-56) located at the KR Puram Access site. The failure domain in question pertains to degradation of SFP/XFP components, fiber cuts, or Rx/Tx power loss. The Arista Access Node operates as a Layer 2/3 switch, employing the eAPI protocol for management and monitoring. The underlying hardware mechanics of the SFP/XFP components are based on optical transceivers that convert electrical signals into optical signals for transmission over fiber optic cables. The Rx/Tx power loss failure mode is a direct consequence of the degradation or physical cuts to the fiber optic cables, leading to a loss of signal strength and ultimately, network connectivity. The eAPI protocol is used to manage and monitor the Arista Access Node, facilitating remote access and configuration. The CLI dialect specified in this SOP adheres to the Arista eAPI CLI, ensuring consistency and accuracy in the execution of diagnostic and resolution commands.

Pre-requisite Checks

- Check: Verify site access and remote hands availability at KR Puram Access.

- Check: Validate redundant link status to ensure no single point of failure

- Check: Ensure all necessary personnel are aware of the network status and the impact of the failure on network operations.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status: Operational, Description: This command verifies the operational status of the interface, confirming whether the interface is up or down. The expected output indicates that the interface is operational, which helps to isolate the failure domain.

Command: show interface detail, Expected Output: Interface Details: SFP/XFP degradation detected, Description: This command provides detailed information about the interface, including any degradation or issues detected. The expected output identifies SFP/XFP degradation as
```

```
the root cause of the failure.

Command: show log, Expected Output: Log Message: Fiber cut detected, Description: This command displays the system log, which may contain error messages or alerts related to the failure. The expected output indicates that a fiber cut has been detected, which is a significant contributing factor to the failure.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP component, Command: replace interface with-sfp , Description: This step involves replacing the degraded SFP/XFP component with a new one. The command replaces the interface with a new SFP/XFP component, ensuring that the interface is operational and the network is restored.

Step 2: Step: Repair fiber cut, Command: repair fiber cut , Description: In the event of a fiber cut, this step involves repairing the damaged fiber optic cable. The command repairs the fiber cut at the specified location, restoring network connectivity.

Step 3: Step: Adjust Rx/Tx power settings, Command: set interface rx-power-adjust , Description: If Rx/Tx power loss is the root cause of the failure, this step involves adjusting the receive and transmit power settings to ensure optimal signal strength. The command adjusts the receive power to the specified level, ensuring that the signal is strong enough to maintain network connectivity.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface Status: Operational, Description: This command verifies that the interface is operational and functioning correctly after the resolution steps have been completed.

Command: show interface detail, Expected Output: Interface Details: No issues detected, Description: This command provides detailed information about the interface and confirms that there are no issues or degradation detected after the resolution steps have been completed.

Command: show log, Expected Output: Log Message: No error messages or alerts, Description: This command displays the system log and verifies that there are no error messages or alerts related to the failure, indicating that the network is stable and functioning correctly.
```

Roll-back Procedures

- Step: Restore original SFP/XFP component, Command: replace interface with-sfp , Description: If the resolution steps fail, this step involves restoring the original SFP/XFP component. The command replaces the interface with the original SFP/XFP component, reverting to the previous configuration.

- Step: Repair original fiber optic cable, Command: repair fiber cut , Description: If the resolution steps fail, this step involves repairing the original fiber optic cable. The command repairs the original fiber cut at the specified location, restoring the original configuration.